Reproducing WLasso on breast cancer gene expression data

Mariana Garcia Mejia

October 23, 2026

0.1 Reproducing WLasso on breast cancer gene expression data (GSE2990)

0.1.1 Data and preprocessing

We use data from the Gene Expression Omnibus with GEO Accession GSE2990 as used by the authors (Zhu et al., 2020) of the paper A variable selection approach for highly correlated predictors in high-dimensional genomic data. The data contains $n = 189$ tumours and 22,283 probes (HG-U133A array: chip to measure human gene expression). The values range from 2.66 to 15.13, which is consistent with RMA-normalised data, meaning that the data is already processed.

```
# Load the gene data from Gene Expression Omnibus
expr <- read.delim("GSE2990_series_matrix.txt",
                   comment.char = "!", row.names = 1)
dim(expr)
```

```
[1] 22283   189
```

```
# Values appear to be normalized (between 2 and 15).
range(as.matrix(expr), na.rm = TRUE)
```

```
[1]  2.656036 15.129603
```

As in Zhu et al. (2021), we kept probes with IQR ≥ 1.5 (1,155 probes) and removed probes without a gene annotation (49 removed, 1,106 left).

```r
# First cleaning step: remove probes with IQR < 1.5
iqr_values <- apply(expr, 1, IQR)
# summary(iqr_values)
keep <- iqr_values >= 1.5
# sum(keep)
expr_filt <- expr[keep, ]
# dim(expr_filt)

# Second step: Remove genes without gene annotation
# BiocManager::install("hgu133a.db")
library(hgu133a.db)

probe_ids <- rownames(expr_filt)
symbols <- mapIds(hgu133a.db,
                  keys = probe_ids,
                  column = "SYMBOL",
                  keytype = "PROBEID",
                  multiVals = "first")

# sum(is.na(symbols))

# Remove
has_symbol <- !is.na(symbols)
expr_ann <- expr_filt[has_symbol, ]
symbols_ann <- symbols[has_symbol]

dim(expr_ann)
```

```
[1] 1106  189
```

The ESR1 probe is a gene associated with breast cancer. We use it as the response y and the remaining 1,105 probes as predictors X. As is convention in implementing Lasso regularizaiton, we standardized each predictor. We also mean-centred y. The paper reports $p = 1{,}111$, so our probe set differs slightly, $p = 1{,}106$.

```r
# Use ESR1 gene as the gene to predict.
all_symbols <- mapIds(hgu133a.db,
                      keys = rownames(expr),
                      column = "SYMBOL",
                      keytype = "PROBEID",
                      multiVals = "first")
```

```r
esr1_probes <- names(all_symbols)[all_symbols %in% "ESR1"]
# esr1_probes
# esr1_probes %in% rownames(expr_ann)

# Pull the ESR Variable left after IQR + Annotation filter and use as y
esr1_probe <- esr1_probes[1]
# esr1_probe
y <- as.numeric(expr_ann[esr1_probe, ])
expr_X <- expr_ann[rownames(expr_ann) != esr1_probe, ]

# length(y)
# dim(expr_X)

# Standardize X.
X <- scale(t(expr_X))
# dim(X)
# round(colMeans(X)[1:3], 6)
# apply(X, 2, sd)[1:3]

# Mean-center Y.
y_c <- y - mean(y)
# mean(y_c) # should be close to zero.
```

0.1.2 Method

The model is $y = X\beta + \epsilon$ with $\mathrm{Cor}(X) = \Sigma$. WLasso whitens the predictors with $\tilde{X} = X\Sigma^{-1/2}$, so that $y = \tilde{X}\tilde{\beta} + \epsilon$ with $\tilde{\beta} = \Sigma^{1/2}\beta$, and the error term is unchanged. It then minimises the generalised Lasso criterion

$$\|y - \tilde{X}\tilde{\beta}\|_2^2 + \lambda\|\Sigma^{-1/2}\tilde{\beta}\|_1,$$

and thresholds the estimates to the top K (for $\tilde{\beta}$) and top M (for β) coefficients, with K and M chosen by an **MSE-ratio rule** controlled by γ. Σ was estimated with the two-block structure of Section 2.5: $(\hat{\alpha}_1, \hat{\alpha}_2, \hat{\alpha}_3) = (0.19, 0.27, 0.55)$, against $(0.17, 0.21, 0.52)$ in the paper.

The paremeters the model takes are γ, which we set to 0.95 (between 0.9 and 0.99 recommended by the paper), and max_steps = 500 (the paper does not find significance difference in performance between 500 and 2000).

0.1.3 A problem with the package and our patch

With the `WLasso` package (v1.0), `Whitening_Lasso()` returned an empty model on these data (the lowest MSE was at the first, all-zero step), although a plain Lasso on the same data selected 60 probes. The package works on its own example data. I used Claude Code to help me identify the issue: using the source, in the step that chooses M, the MSE is computed from the coefficients sorted by absolute value, not from the coefficients in their original order, so each coefficient multiplies the wrong column of X. We think this is a bug that does little harm when all correlations are positive, as in the paper's simulations. We changed that one argument in a copy of the function (not an official fix), and results below come from that modified copy.

```
# Install WLasso package
# install.packages("WLasso")
library(WLasso)
# help(package = "WLasso")
```

```
# Step 1: Sigma Estimation
Sigma_est <- Sigma_Estimation(X)
round(Sigma_est$alpha, 2) # paper reports 0.17, 0.21 and 0.52, we get slightly different
```

```
[1] 0.19 0.27 0.55
```

```
# We get: 0.19 0.27 0.55

# Number of probes the clustering put into one of the two blocks
# We get 181 out of 1105
length(Sigma_est$group_act)
```

```
[1] 181
```

```
# Step 2: Model transformation
Sigma_estmat <- Sigma_est$mat
```

```
# Fix bug on the paper
src <- paste(deparse(WLasso::Whitening_Lasso), collapse = "\n")
src2 <- gsub("x\\s*=\\s*beta_sort,\\s*sorted_vect\\s*=\\s*beta_sort",
             "x = beta_final0, sorted_vect = beta_sort", src)
```

```r
# identical(src, src2)   # we want FALSE this time

WL_fixed <- eval(parse(text = src2))
environment(WL_fixed) <- asNamespace("WLasso")
# grep("beta_sort", strsplit(src, "\n")[[1]], value = TRUE)
# grep("beta_interm", strsplit(src2, "\n")[[1]], value = TRUE)
```

```r
# Fit the model
# mod_fix <- WL_fixed(X = X, Y = y_c, Sigma = Sigma_estmat,
                      # gamma = 0.95, maxsteps = 500)

# range(mod_fix$mse)
# sum(mod_fix$beta.min != 0)
```

```r
# We also fit regular Lasso for comparison:
#install.packages("glmnet")
# library(glmnet)
# cvfit <- cv.glmnet(X, y_c)
# sum(coef(cvfit, s = "lambda.min")[-1] != 0)
# returns 60 coefficients!!!!
```

0.1.4 Results

Method	Probes	Genes
WLasso ($\gamma = 0.95$, patched)	59	56
Lasso (cross-validation)	60	59

0.1.5 Key takeaways

- WLasso Works primarily for quantitative data, there is an MSE minimization that happens when fitting the Lasso.
- The WLasso package presented some errors, so it might be better to fit the math directly rather than relying on the package.
- "WLogit" Package developed by the same authors.

Reference. Zhu, W., Lévy-Leduc, C. and Ternès, N. (2021). A variable selection approach for highly correlated predictors in high-dimensional genomic data. *Bioinformatics*, 37(16), 2238–2244.